COMP 3920 Database Systems 1

MongoDB Atlas

Introduction:

The goal of this document is to serve as a walkthrough for working with MongoDB as a NoSQL database.

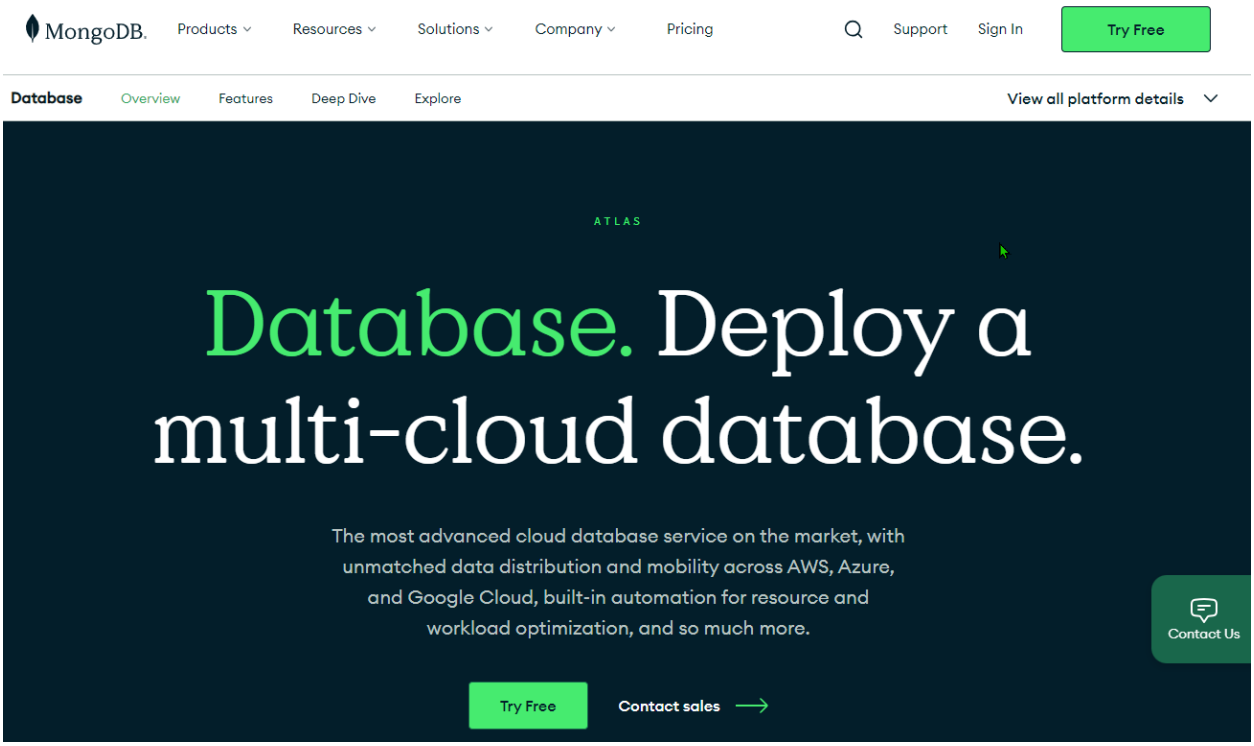
In this lab we will be installing a few tools to get things going. We will be installing:

- **MongoDB Atlas** - this is a cloud service for hosting MongoDB databases. We will create an account to create database collections.
- **Studio 3T** - this is a program to manage MongoDB databases. It can connect to local and remote databases and can perform many of the CRUD operations we need (this is similar to MySQL Workbench)

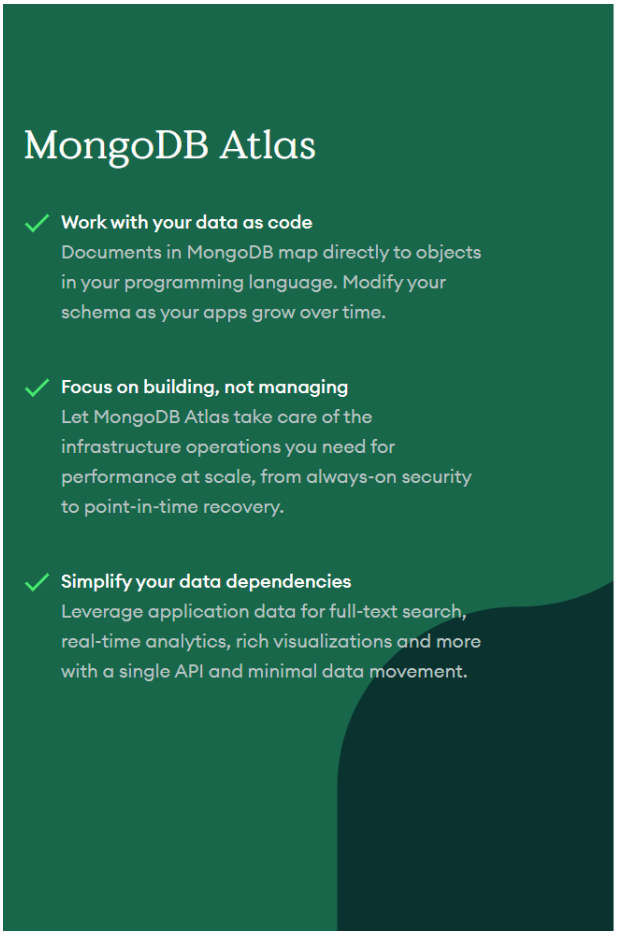
Step 1: Create a MongoDB Atlas Account.

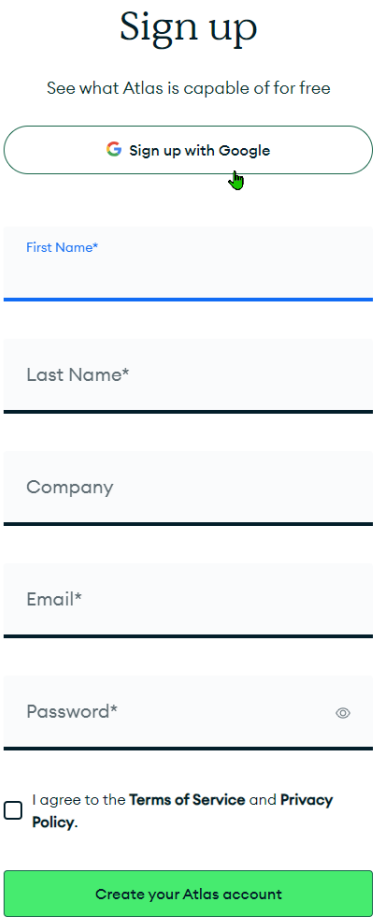
Go to: https://www.mongodb.com/atlas/database

Click "Try Free".

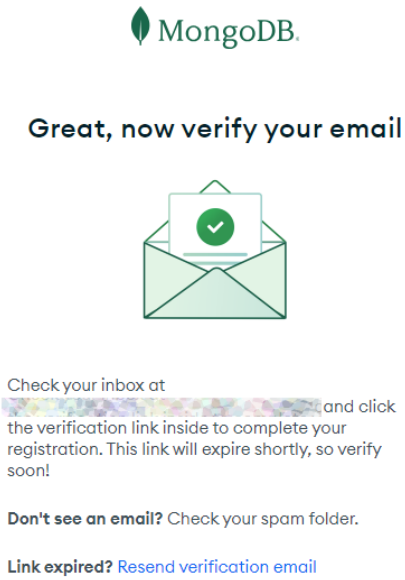


Complete the form to continue.

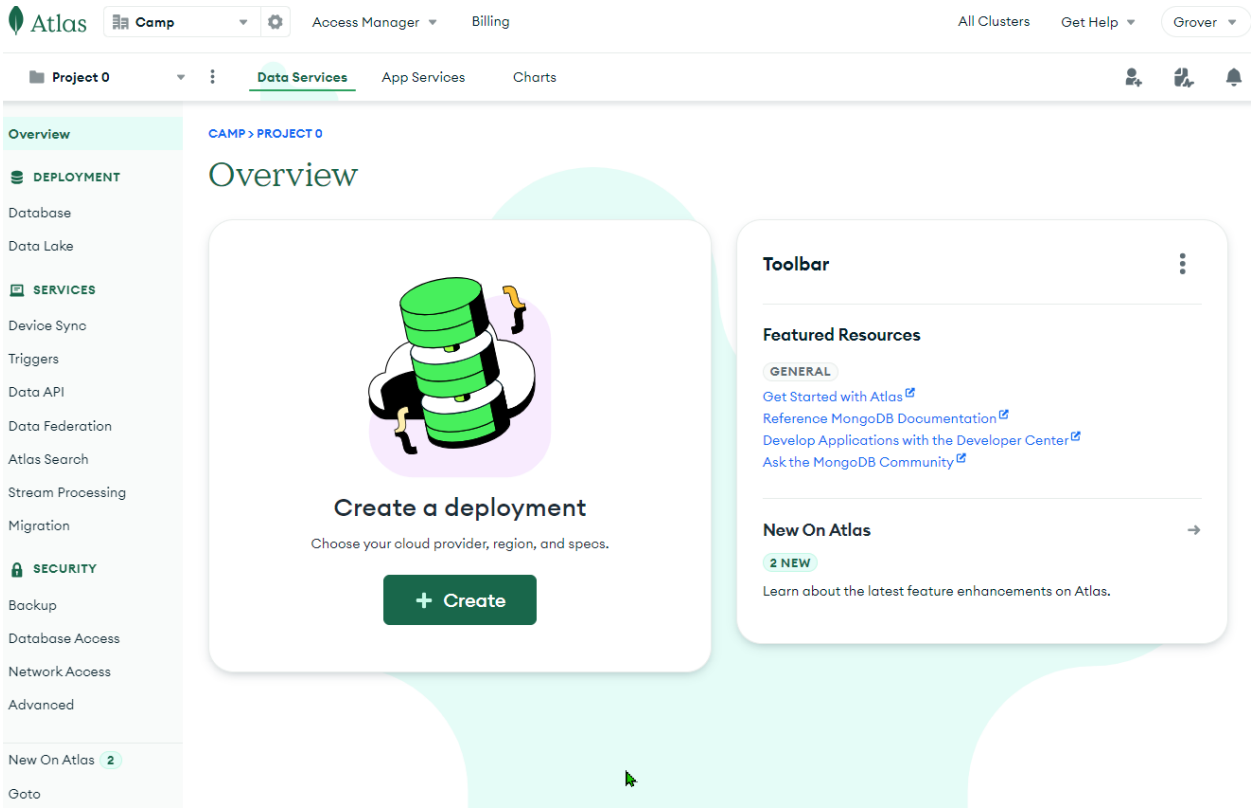




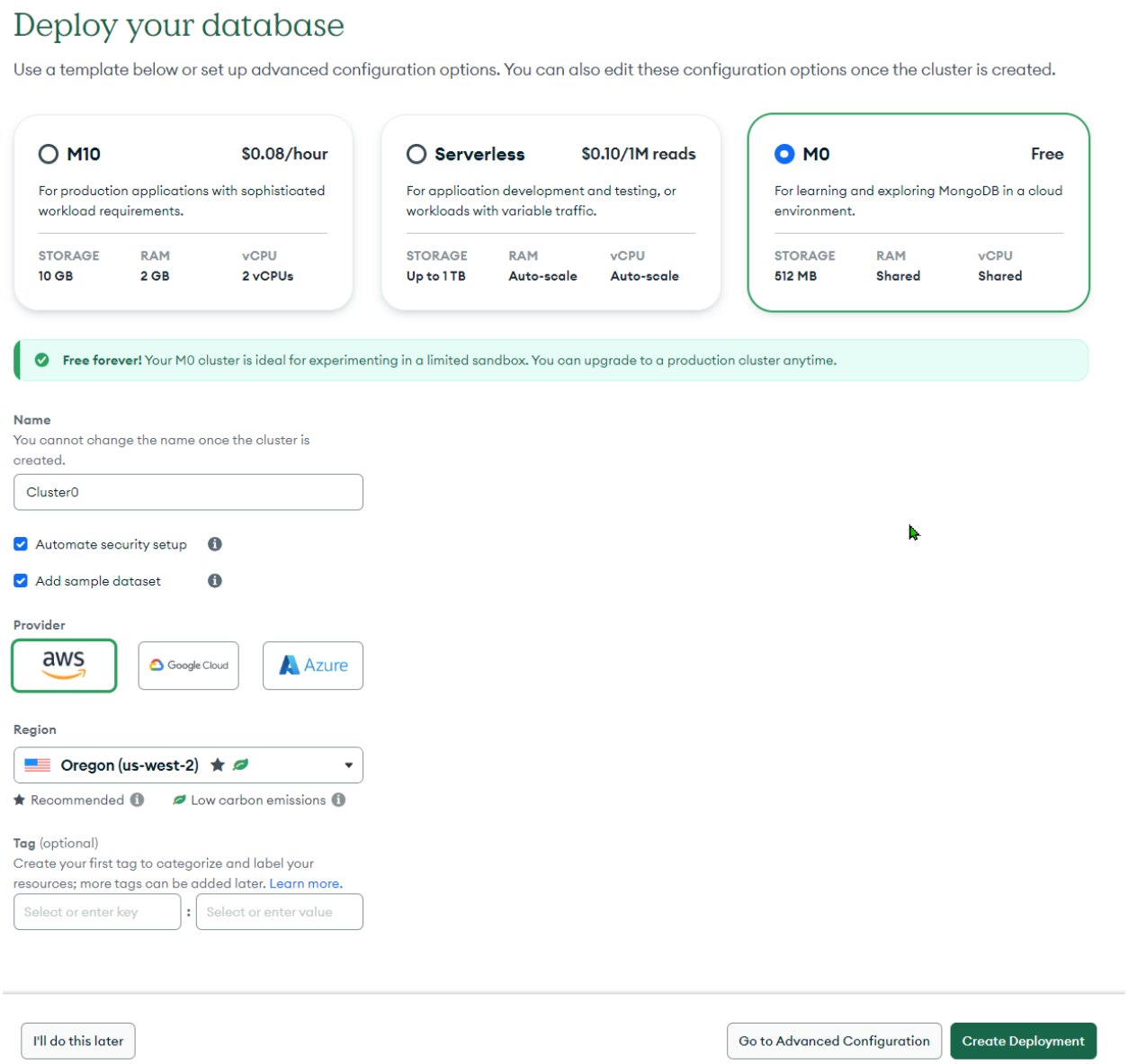
Make sure to verify your email account.



To create a new Deployment, click "Create" under "Create a deployment".

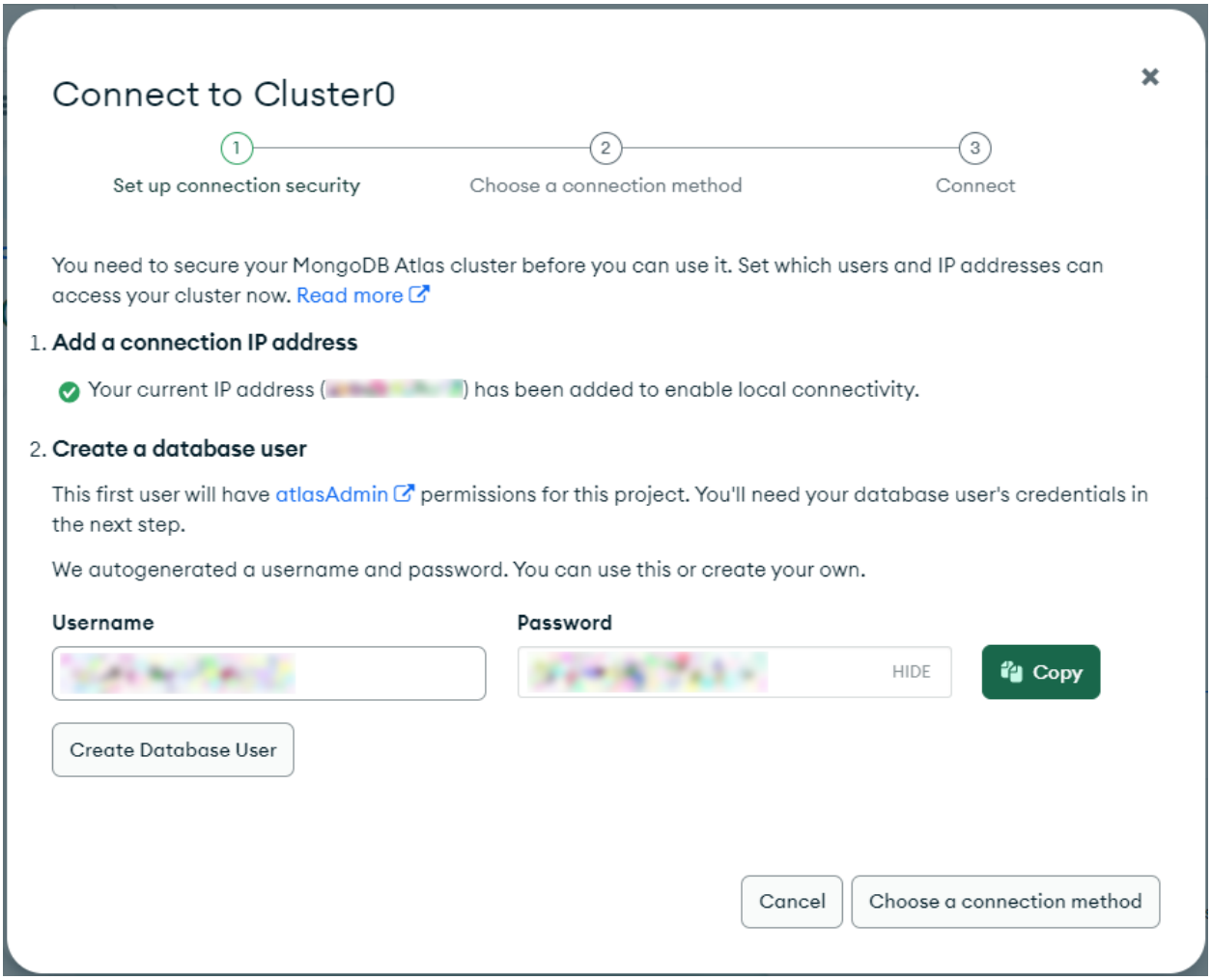


Make sure to pick M0 as your server type! This way you can use the free tier service!



You can leave all the other settings as default.
Click "Create Deployment".

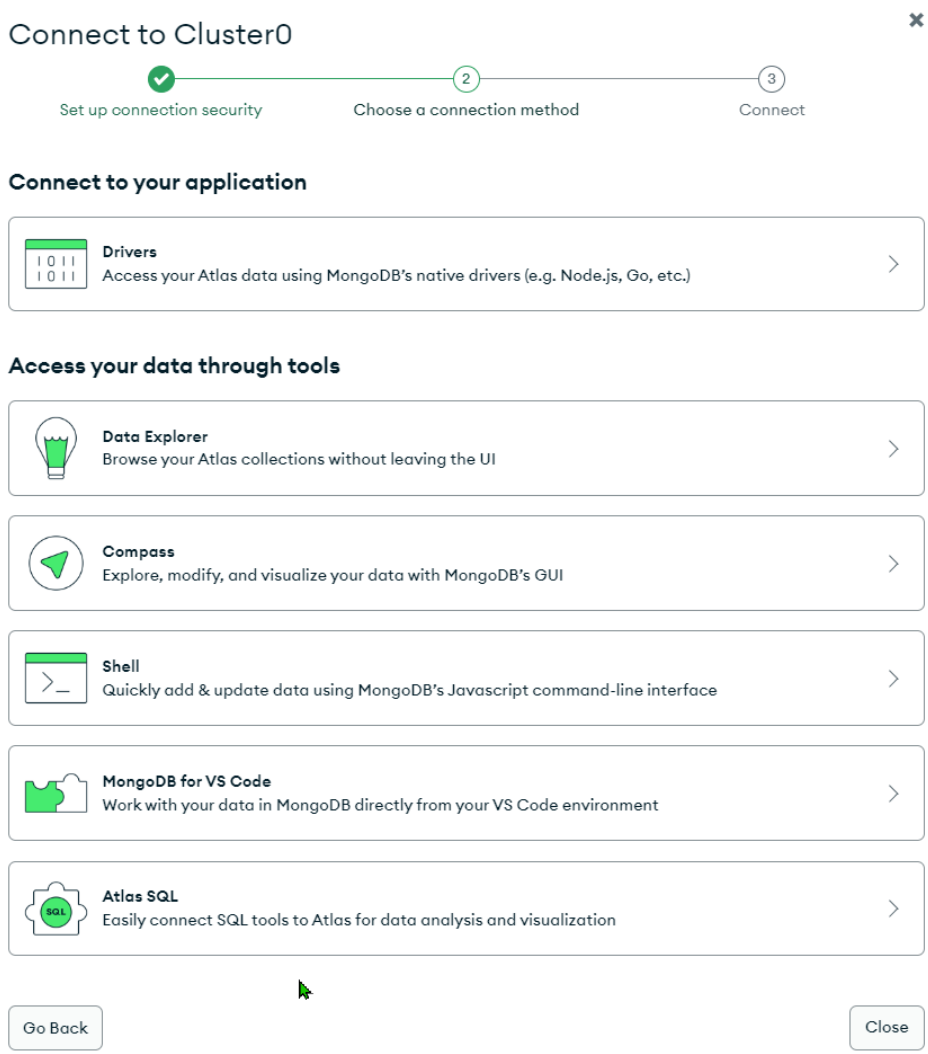
Create a new user and set the password. Remember this password, you'll need this to connect to your database!



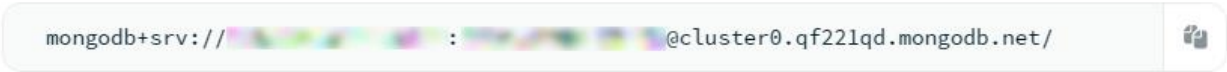
You can use the default password (but make sure to copy it and save it somewhere safe) or create your own.
Click "Create Database User".

Click "Choose a connection method".

Click "MongoDB for VS Code".



Look for a connection string like this:



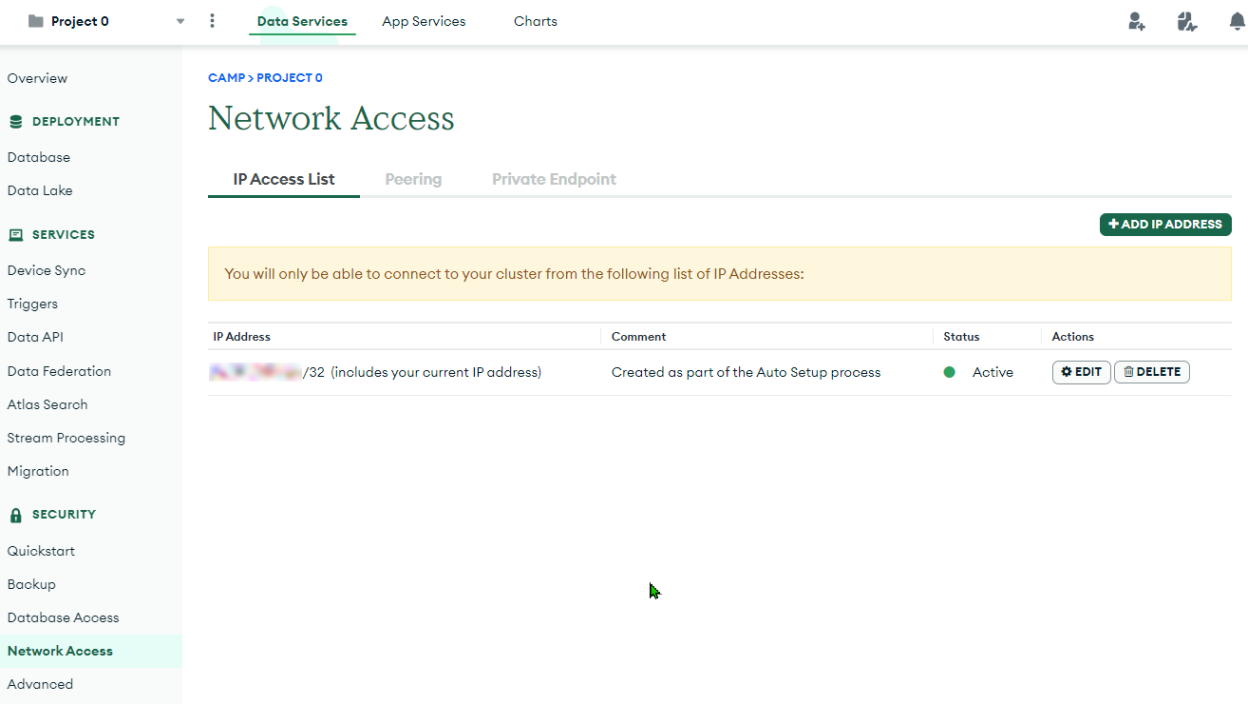
This is what you will need to connect to your database from your code. Copy this string and save it for later.

Click "Review setup steps".

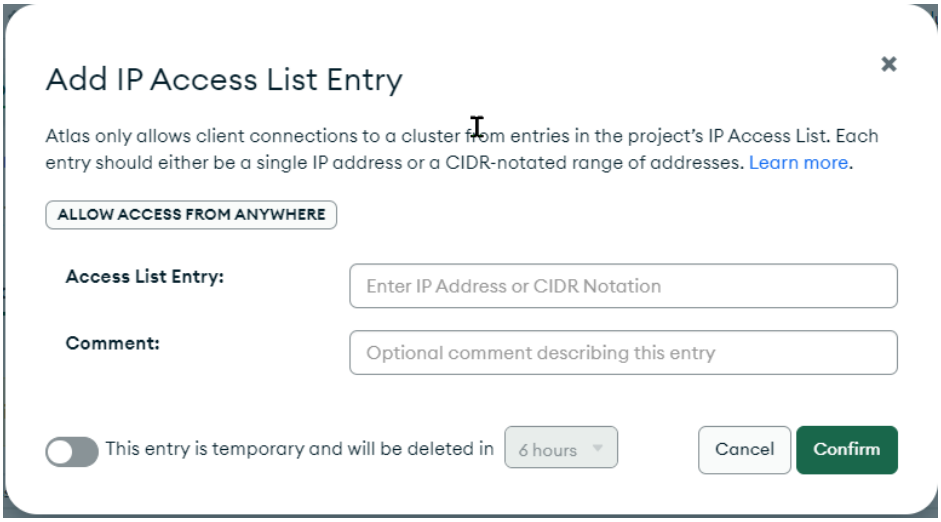
Click "Done".

IMPORTANT! In order for a hosting service like Render to be able to access your MongoDB database, you'll need to add all IP addresses to the list of allowed IPs.

You can click on "Network Access" on the left-hand navigation.

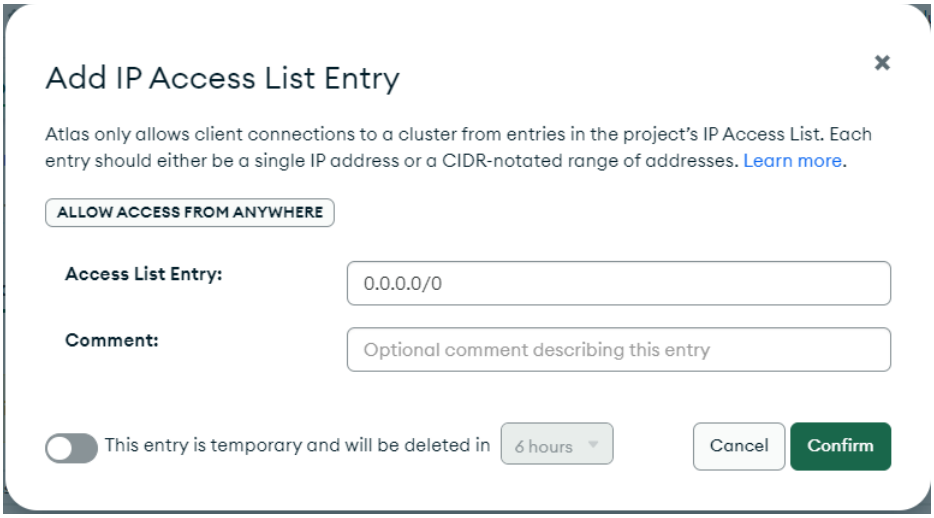


Click "Add IP Address".



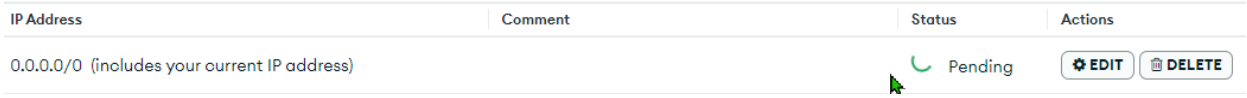
Click "Allow Access From Anywhere".

(this should Add 0.0.0.0/0 to the Access List Entry)

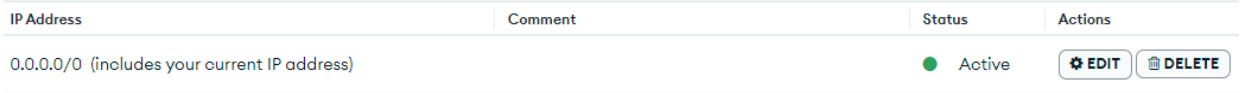


Click "Confirm".

This may take a few moments to take effect.



(wait for the status to go from "Pending" to "Active")

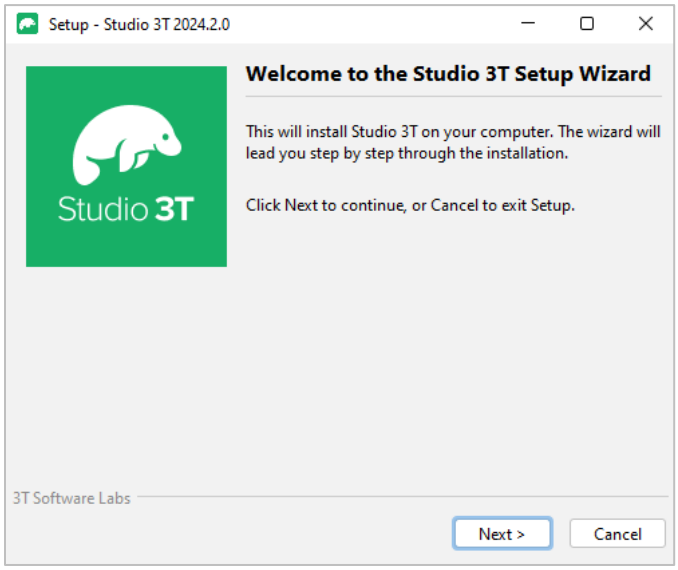


Step 2: Install Studio 3T.

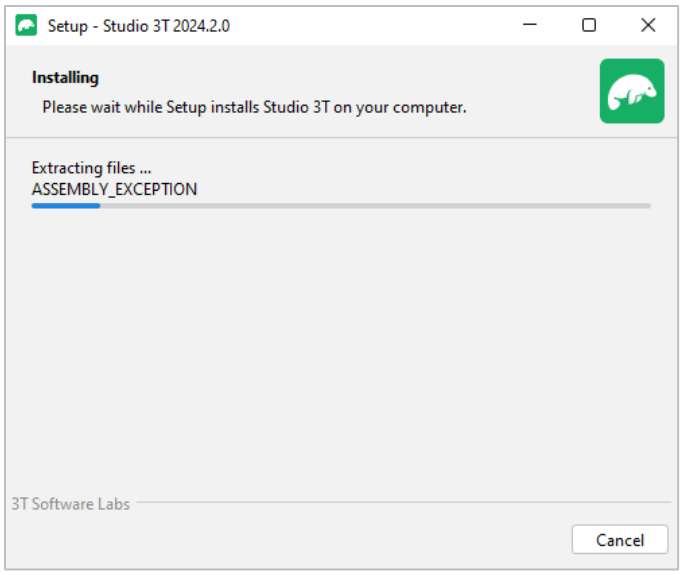
Browse to: https://studio3t.com

Click "Download".

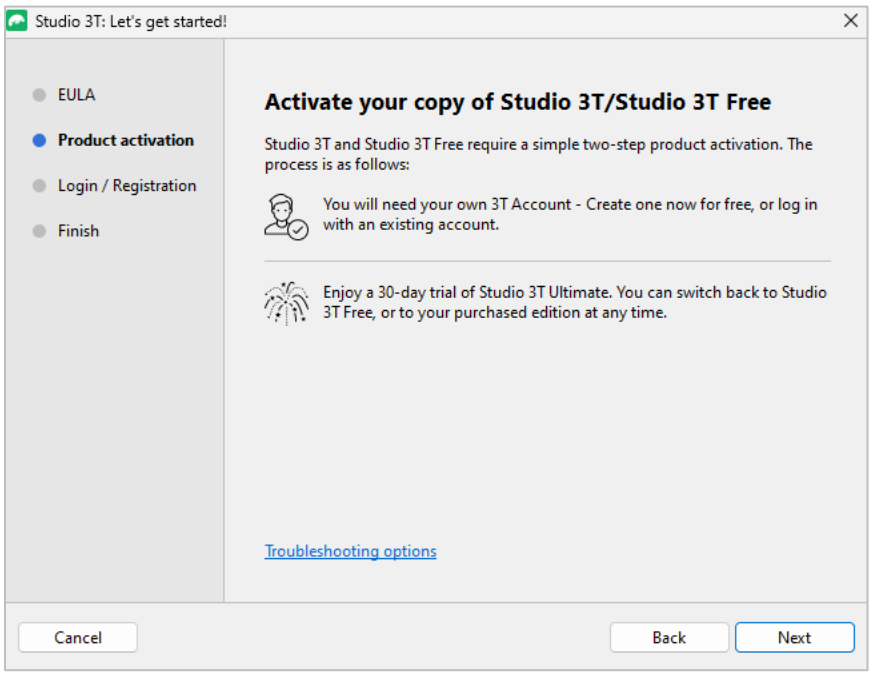
Open and run the software to install.



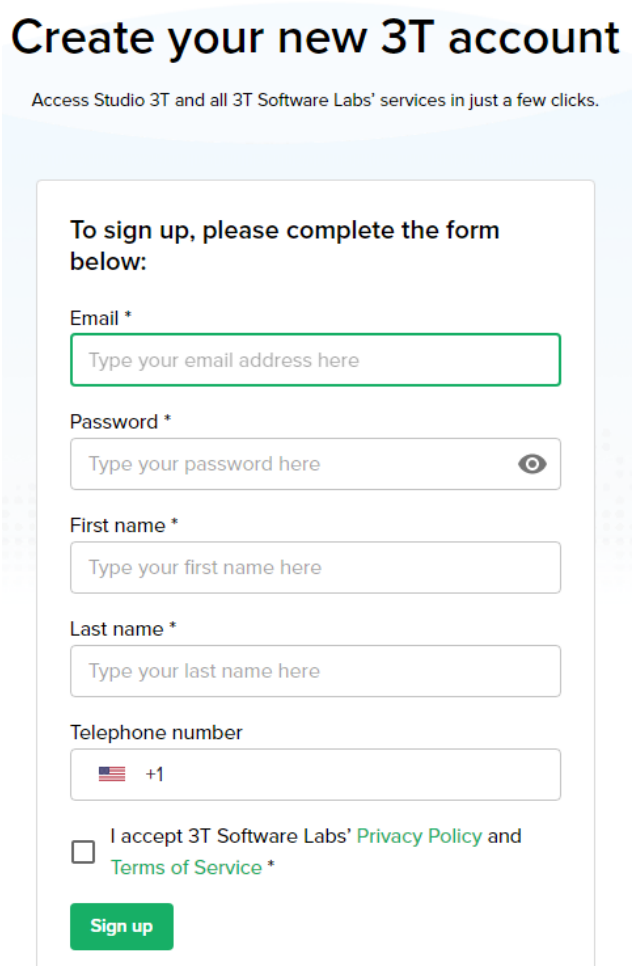
Follow the installer and wait for it to complete.

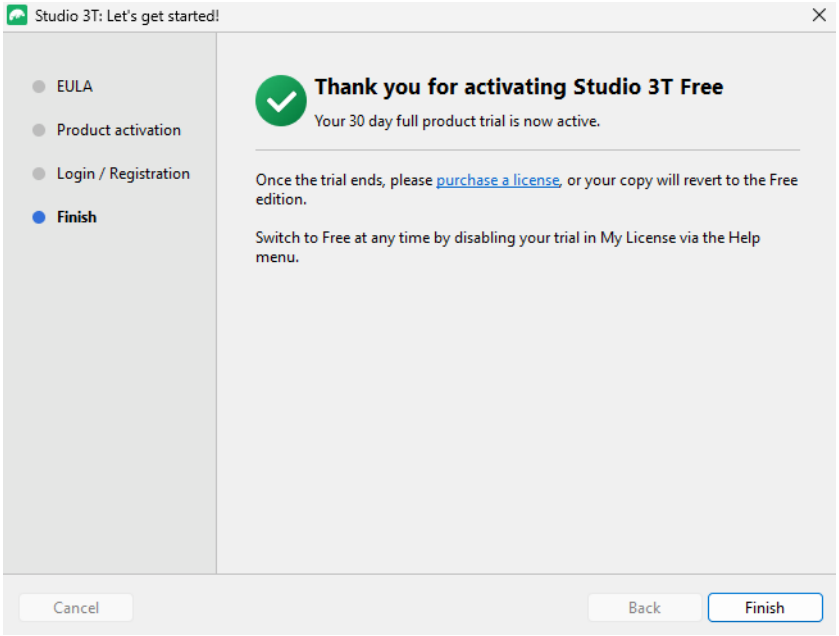


Without a license, Studio 3T will revert to the free version.



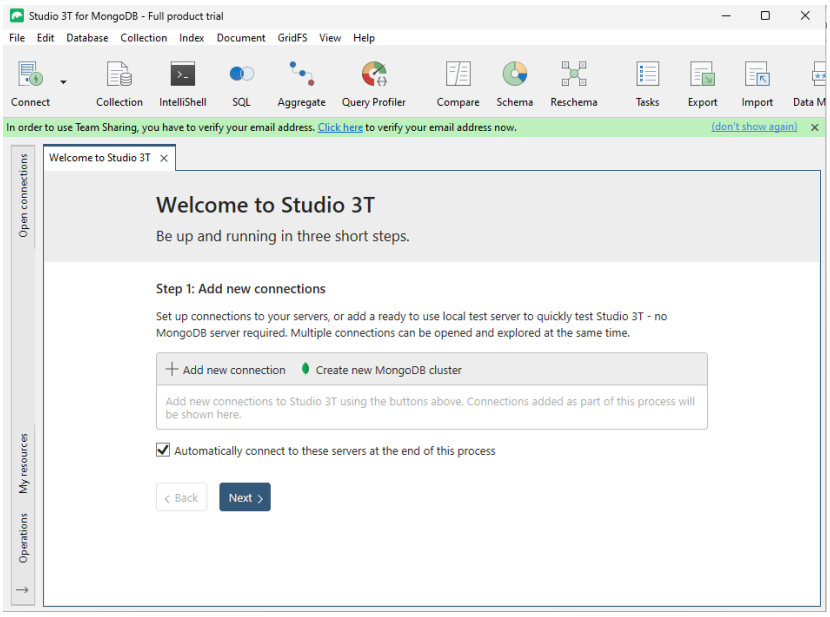
But you will need to create an account:



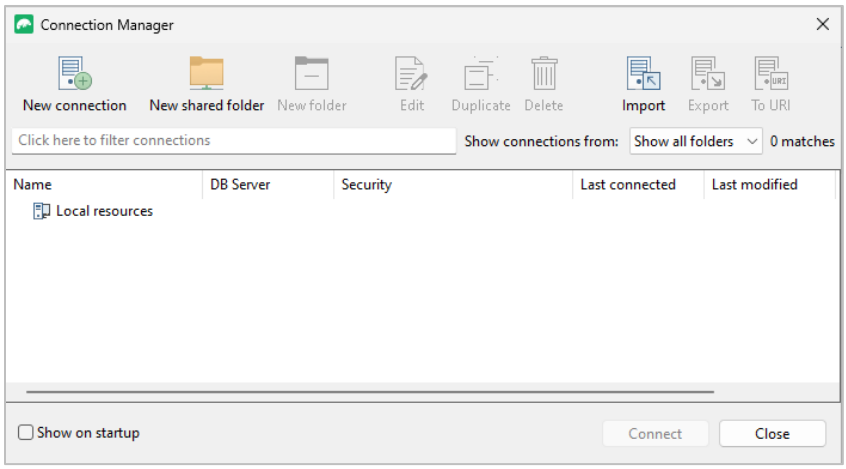


Once the account has been activated, Click "Finish".

From the start up screen click "Connect".

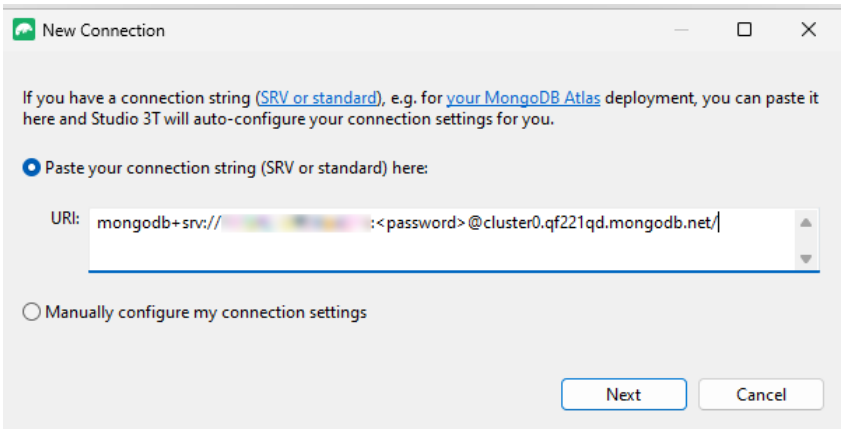


Then click "New Connection".



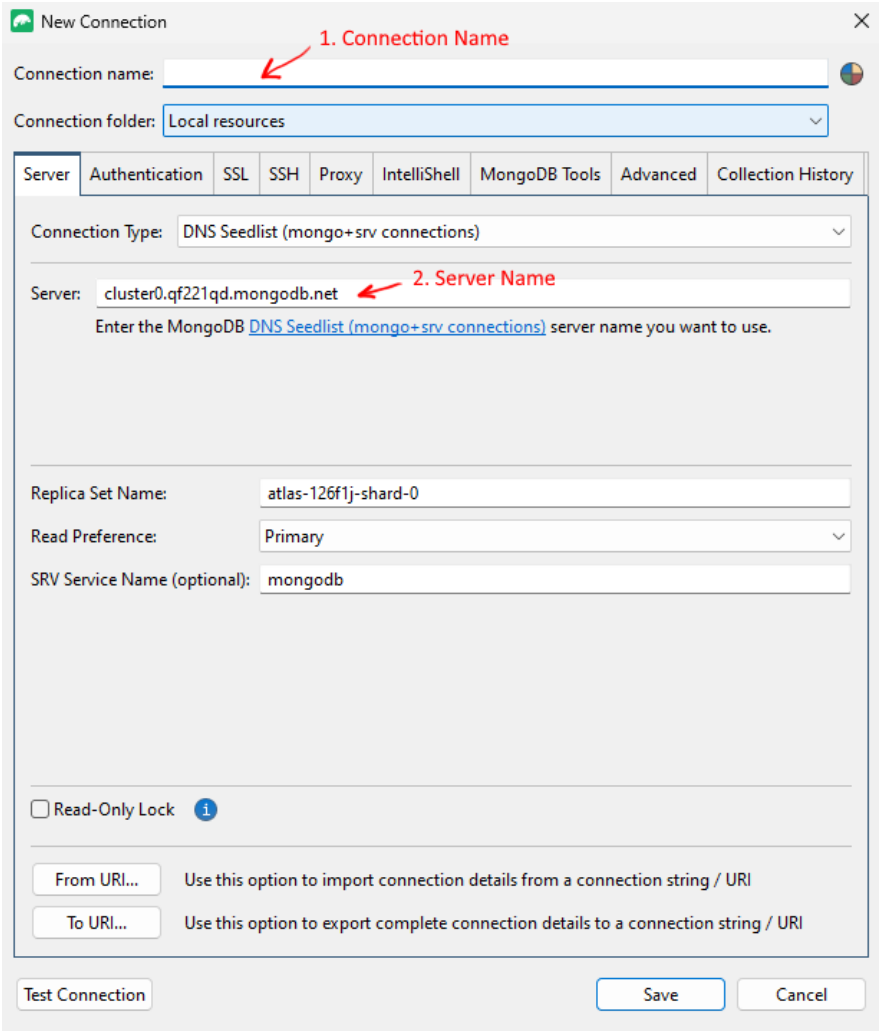
Remember that connection string we got from MongoDB Atlas?

You'll copy/paste it into the URI field:

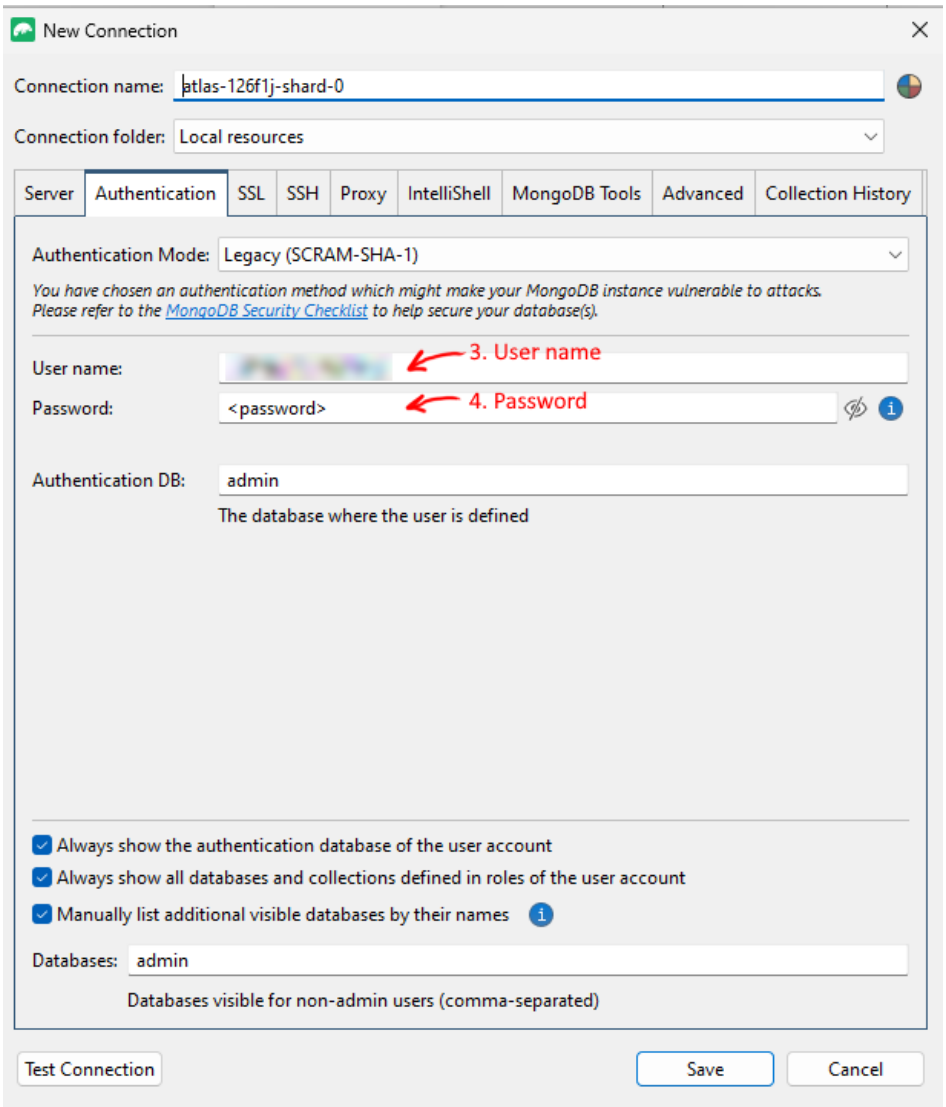


Click "Next".

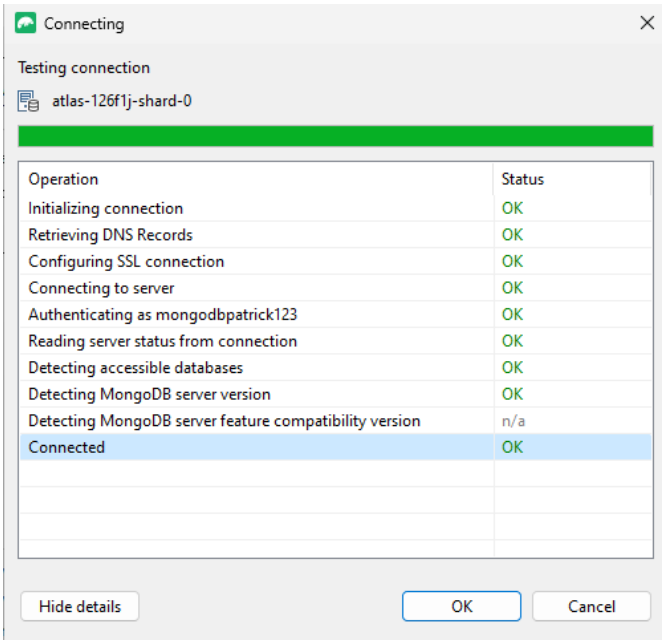
Give your connection a name (ex: MongoDB Atlas) and verify the correct information has been entered:



Verify your user name.
If your connection string contained `<password>` instead your actual password, this is wrong and you need to change it! You'll need to re-enter the correct password:

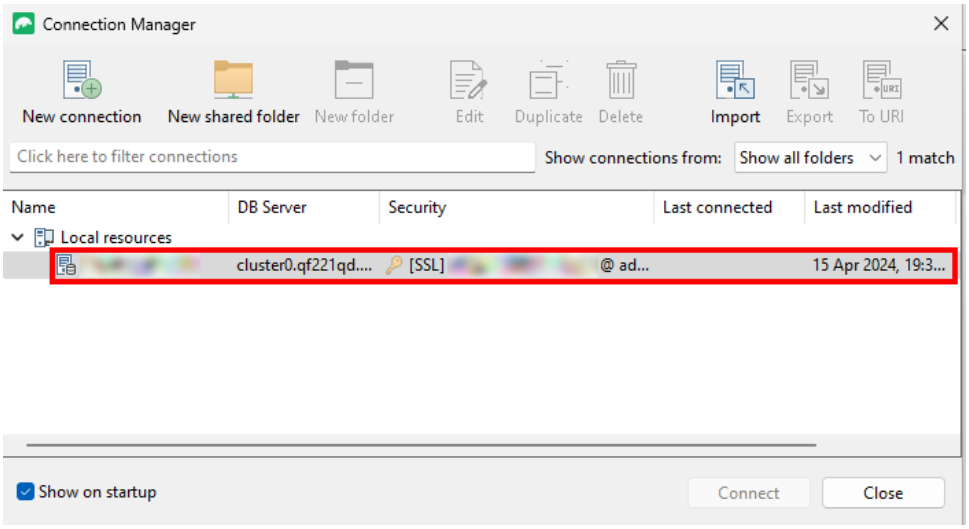


Click "Test Connection".

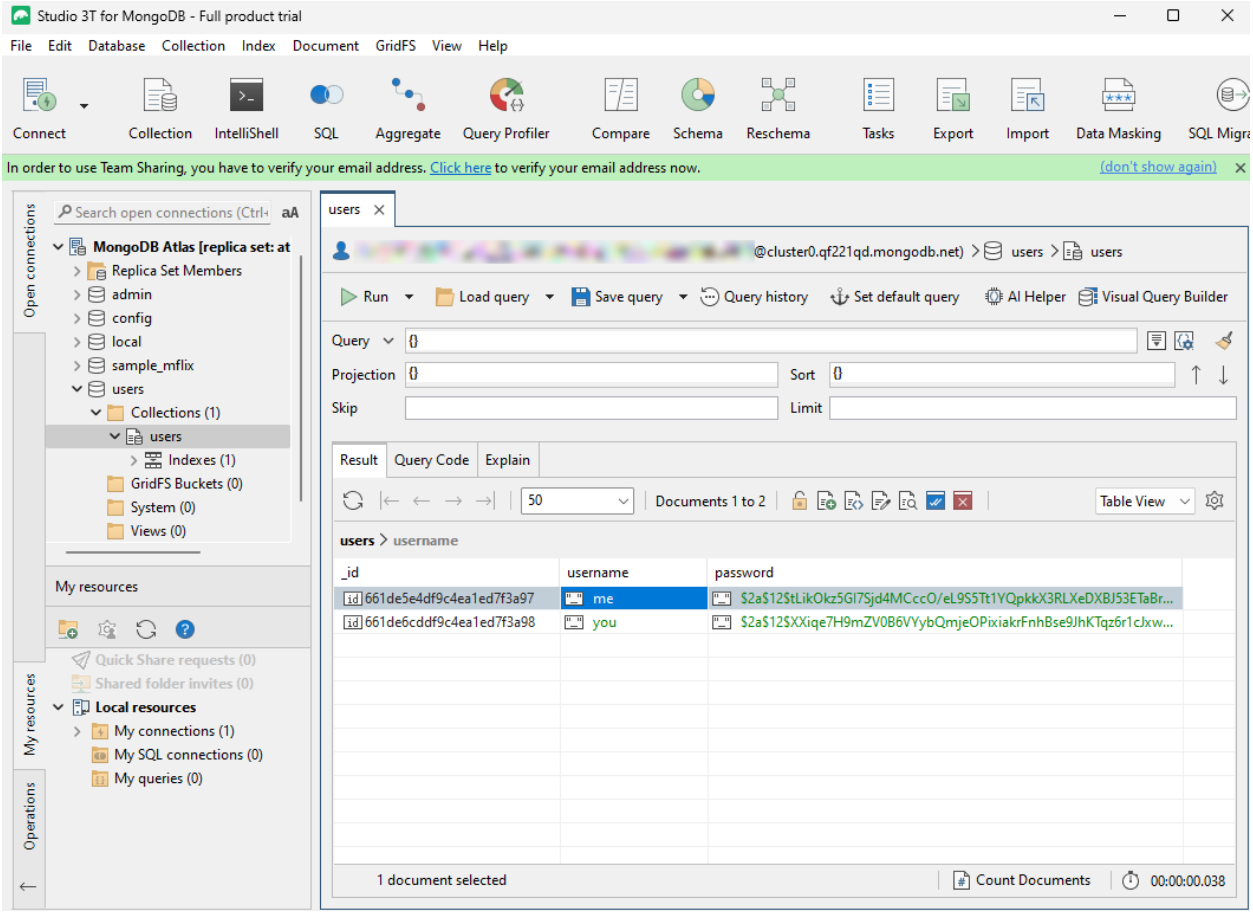


If it worked, you should get all green **OK**s (and possibility an n/a) in the connection test results.

Your new connection should show up in the Connection Manager like this:



Select your connection and click "Connect".


Studio 3T for MongoDB - Full product trial
File Edit Database Collection Index Document GridFS View Help
Connect Collection IntelliShell SQL Aggregate Query Profiler Compare Schema Reschema Tasks Export Import Data Masking SQL Migra
In order to use Team Sharing, you have to verify your email address. Click here to verify your email address now.
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Open connections
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